

A foundational problem

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As you will remember, we resorted to conditioning on $\mathbb{Q}_m = \mathcal{Q} \rightarrow \mathbf{A}_m = a$, instead of $\mathbf{A}(\mathcal{Q}) = a$, because that way the knowledge in $\mathcal{Q}$ (for example, about whether $\mathcal{Q}$ itself or other states satisfy this or that easy to check syntactic property) was used to implement the policy (translate from small to large beliefs). Unexpectedly, it seems like this doesn't achieve what we thought it did.

Say the inductor state $\mathbb{Q}_k$ will be extrapolated, and frozen as a prior $\mathbb{P}$. Adopt the notation $\uparrow x$ for " x is sufficiently high" (cleaner than dragging lower bounds). Say

$$\uparrow \mathbb{Q}_k(\forall \mathcal{Q}(P(\mathcal{Q}) \wedge (\mathbb{Q}_m = \mathcal{Q} \rightarrow \mathbf{A}_m = a) \rightarrow \uparrow U))$$

That is, $\mathbb{Q}_k$ believes strongly that "for any state $\mathcal{Q}$, if it satisfies the property $P()$, and $\mathbf{A}$ takes action a when $\mathcal{Q}$ is the case, then U ¹ is high".

Then, by universal instantiation, for any specific state $\underline{\mathcal{Q}}$, we'll have

$$\uparrow \mathbb{P}(P(\underline{\mathcal{Q}}) \wedge (\mathbb{Q}_m = \underline{\mathcal{Q}} \rightarrow \mathbf{A}_m = a) \rightarrow \uparrow U) \quad (\text{Implication})$$

Say it is actually the case that $P(\underline{\mathcal{Q}})$, and furthermore, that $\underline{\mathcal{Q}}$ is aware of this fact, so that

$$\uparrow \underline{\mathcal{Q}}(P(\underline{\mathcal{Q}}))$$

What we want to happen is that

$$\uparrow \mathbb{P}(\uparrow U \mid \mathbb{Q}_m = \underline{\mathcal{Q}} \rightarrow \mathbf{A}_m = a) \quad (\text{Desire})$$

This will imply

$$\uparrow \mathbb{E}^{\mathbb{P}}(U \mid \mathbb{Q}_m = \underline{\mathcal{Q}} \rightarrow \mathbf{A}_m = a)$$

and so the action a will be chosen, as desired.

So, how do we ensure that (Desire) is satisfied? Well, it will be satisfied if

$$\uparrow \mathbb{P}(P(\underline{\mathcal{Q}}) \mid \mathbb{Q}_m = \underline{\mathcal{Q}} \rightarrow \mathbf{A}_m = a)$$

because then we can use (Implication) with propositional coherence. But how do we ensure that? When conditioning on $\mathbb{Q}_m = \underline{\mathcal{Q}} \rightarrow \mathbf{A}_m = a$, we restrict $\mathbb{P}$ to two kinds of worlds: those in which $\neg \mathbb{Q}_m = \underline{\mathcal{Q}}$, and those in which $\mathbb{Q}_m = \underline{\mathcal{Q}} \wedge \mathbf{A}_m = a$. It is true that in the latter worlds we will have high probability on $P(\underline{\mathcal{Q}})$, by BLI self-trust. But in the former this might not be the case. And

¹Recall that U is global time-penalized utility, that is, not restricted to the worlds in which $\mathcal{Q}$ is actually the case.

indeed, if $\mathbb{Q}_m = \underline{\mathcal{Q}}$ is a priori unlikely enough according to $\mathbb{P}$ (so that the former worlds are a large enough fraction), and in enough of those former worlds $P(\underline{\mathcal{Q}})$ is false, we won't get what we need.

Another way to frame this failure is that we do get $\uparrow \mathbb{P}(\mathbb{Q}_m = \underline{\mathcal{Q}} \rightarrow (\mathbf{A}_m = a \rightarrow \uparrow U))$, but this doesn't imply that $\uparrow \mathbb{P}(\uparrow U \mid \mathbb{Q}_m = \underline{\mathcal{Q}} \rightarrow \mathbf{A}_m = a)$. It does imply $\uparrow \mathbb{P}(\uparrow U \mid \mathbb{Q}_m = \underline{\mathcal{Q}} \wedge \mathbf{A}_m = a)$, but of course that's just being completely updateful (which we already knew solved this problem, but at the expense of updatelessness).

This is my interpretation of the problem:

We thought that, somehow, when considering the implication the knowledge inside $\underline{\mathcal{Q}}$ was only used "in the right places". That is, the knowledge that $P(\underline{\mathcal{Q}})$ is true was transmitted, but somehow the other knowledge of $\underline{\mathcal{Q}}$ wasn't used everywhere (in which case we'd be completely updateful).

But what was actually going on, unbeknownst to us, is that in a few places the knowledge of $\underline{\mathcal{Q}}$ was being used completely (the worlds with $\mathbb{Q}_m = \underline{\mathcal{Q}} \wedge \mathbf{A}_m = a$), and in others it wasn't being used at all (the worlds with $\neg \mathbb{Q}_m = \underline{\mathcal{Q}}$), and the latter is why the above argument doesn't go through (contrary to what we thought). Indeed, we face again the same problem as when conditioning on $\mathbf{A}(\underline{\mathcal{Q}}) = a$: how do we ensure the knowledge about $P(\mathbb{Q}_m = \underline{\mathcal{Q}})$ is used, but not the rest? If we don't condition on anything (as in the $\neg \mathbb{Q}_m = \underline{\mathcal{Q}}$ worlds above), the knowledge is not used. If we condition on the whole $\mathbb{Q}_m = \underline{\mathcal{Q}}$ (as in the $\mathbb{Q}_m = \underline{\mathcal{Q}} \wedge \mathbf{A}_m = a$ worlds above), we are completely updateful.